# C PROGRAMMING — ASSIGNMENT 1

## Task 10: Compare Expectations with Execution

Marks: 4 | Total assignment: 50 marks

### Scale calculations

To compare types, record totals from actual program runs for 4,000, 60,000 and 800,000 sample bills. A simple transparent experiment is to add the same integer-paise bill N times and compare it with float rupee addition. Do not report machine-specific observed values without running on your system; use the included experiment and attach its output.

### Overflow

The C standard guarantees minimum ranges, but sizeof and exact ranges vary by implementation. int and long may be 32 bits on common systems; long long is at least 64 bits. At 800,000 bills, a 64-bit signed integer paise total can represent very large sums, but whether int or long overflows depends on the average bill and platform. Check before addition if totals can approach LLONG_MAX.

### Double versus long long

A double usually has a very large exponent range and is unlikely to overflow for 800,000 ordinary bills, but it is approximate and may not represent every paise exactly. long long represents integer paise exactly until overflow. Both additions are O(1) and each variable uses constant space; sizeof(double) and sizeof(long long) should be measured.

### C Source Code / Implementation

```
#include <stdio.h>
#include <limits.h>
#include <float.h>
#include <stdint.h>

int main(void) {
    const long long bill_paise = 16250LL; /* Rs 162.50 */
    const int counts[] = {4000, 60000, 800000};
    printf("sizeof(int)=%zu sizeof(long)=%zu sizeof(long long)=%zu sizeof(double)=%zu\n",
           sizeof(int), sizeof(long), sizeof(long long), sizeof(double));
    printf("INT_MAX=%d LONG_MAX=%ld LLONG_MAX=%lld DBL_DIG=%d\n",
           INT_MAX, LONG_MAX, LLONG_MAX, DBL_DIG);
    for (size_t k = 0; k < sizeof counts / sizeof counts[0]; ++k) {
        long long exact = (long long)counts[k] * bill_paise;
        double approximate = (double)counts[k] * 162.50;
        printf("%d bills: integer=%lld paise (Rs %lld.%02lld), double=%.2f rupees\n",
               counts[k], exact, exact / 100, exact % 100, approximate);
    }
    return 0;
}
```

### Submission note

Compile and run the code on your own system, save actual terminal output/screenshots, and include them in your report. Review the code and be prepared to explain it during the viva. Any machine-specific measurements must come from your own execution.